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Fast and Robust In-Situ Quantitative Mineral Analysis of Carbonate Rocks with AI-Driven Hyperspectral Unmixing Approach

I. Qudsi, A. Herlambang, and A. Koeshidayatullah, College of Petroleum Engineering and Geosciences, King Fahd University of Petroleum and Minerals, Dhahran, Saudi Arabia

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Abstract

Hyperspectral sensor measured the reflected electromagnetic wave within visible, near, middle, and long-wave infrared intervals. In geology, Short-wave Infrared (SWIR) and Long-wave Infrared (LWIR) wavelength ranges have been proven to be useful to detect mineralogy variation of a material. The purest material is usually unable to be captured in many cases, especially in distance measurements such as airborne or satellite. Each measurement, a single pixel in an image or point, usually captures the spectra from the mixture of several minerals. Spectral unmixing is a method to estimate the abundance of constituent minerals and their distribution in the mixed material of an object being measured. Using endmembers retrieved from the purest endmember measured or spectral libraries, the relative composition (abundance) of every material is quantified by analyzing the influence of the endmember's spectra on the noisy/impure spectra of the given mixed material. In this project, the abundance of Calcite, Dolomite, and Quartz of rock powder samples from carbonate outcrops was quantified using several techniques, including linear, PLSR, and machine learning assisted hyperspectral unmixing methods, and the results were compared with other conventional determination methods. The powder samples are measured using a handheld hyperspectral measurement device, a spectroradiometer, that has a 2 cm measured surface. XRD measurement was used to validate the mineral abundance quantification performed by every method applied to the measured spectra. The result of this study highlights the potential applications of a portable spectroradiometer as initial quantitative mineral analysis, leveraging the fast and non-destructive data acquisition supported by the AI unmixing algorithm for abundance estimation.

Introduction

Hyperspectral sensors have more spectral bands compared to multispectral, making them useful for discriminating objects that have similar reflectance spectra (Zhenfeng Shao et al. 2014). In geologic remote sensing, it elevates the detail of mineral identification from material bearing classes to quantitatively extract the mineralogy of the outcrop or rock samples (van der Meer et al. 2012; Peyghambari and Zhang 2021). The reflectance spectra of minerals with similar composition tend to have adjacent wavelength's location of the absorption feature signatures, thus multispectral in many cases could not solve this issue. Leveraging the

additional channels in the hyperspectral sensor makes it possible to further analyze the effect of processes that create the absorption features like electronic and vibrational processes, the amount of light scattered and absorbed by different grain sizes, etc.

The solar spectral range, Visible-Near Infrared (VNIR) and Short-Wave Infrared (SWIR), within the range of 400-2500 nm, provides information about plenty of valuable minerals on earth, where VNIR is usually related to the surface roughness and iron-bearing minerals, while SWIR features the hydroxyl-bearing minerals, sulfates, carbonate, and many hydrothermal alteration and other geological features (Clark et al. 1990). In many geological study cases, the combination of VNIR and SWIR sensors is usually enough to extract the required information for the identification of hand-specimen samples, remote sensing, and core images. This method provides rapid analysis with little to no destruction required to be applied to the rock samples, which is important to preserve the rock for further analysis.

Despite its ability to distinguish the signature of the reflectance spectra from various minerals, one of the biggest issues in hyperspectral geology studies is the mixing of spectra. The spectra of minerals are often observed to be complex in relation to the scattering process, grain size, abundance, and their opacity (Clark et al. 1990; Zhang et al. 2016; Dixit and Agarwal 2021). The pixel resolutions of most satellites and airborne sensors are too large to capture any surface of homogeneous minerals. However, this could also happen even in the higher resolution of imagery, such as laboratory-recorded Hyperspectral Image (HSI).

In geology, it is very common to have a spectral mixture even in a higher spatial resolution image with a millimeter pixel size (Fig. 1), especially when dealing with sedimentary rocks that have <2 mm grain sizes. While the spectral resolution of the hyperspectral is able to distinguish the signature spectra of clay minerals, this will be recorded as mixed spectra eventually due to this issue. Therefore, unmixing techniques play vital roles in extracting the information to the fullest potential from hyperspectral images and point measurements of the rock samples, including understanding the abundance of mineral composition of the mixed spectra (Sabins 1999; Zhang et al. 2016; Scafutto et al. 2017; Tuşa et al. 2020).

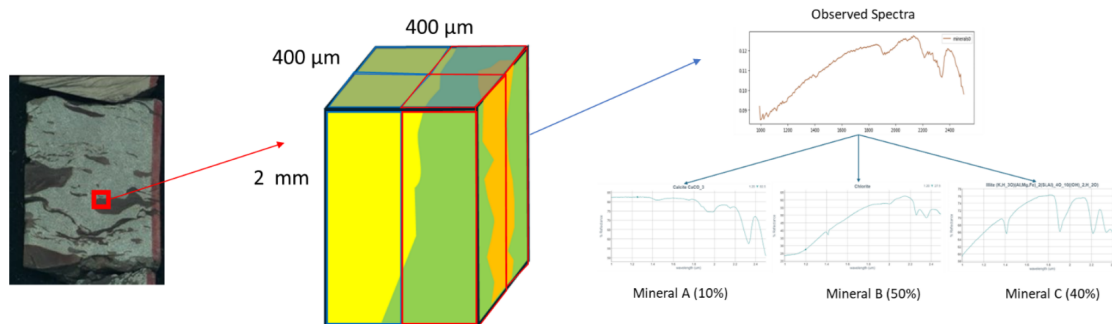


Figure 1—Illustration of the mineral composition of mixed observed spectra from a 400×400 µm pixel size from the HSI of drill-core

Aside from the most conventional Linear Spectral Unmixing, attempts to incorporate artificial intelligence in performing non-linear unmixing are gaining popularity in recent years to improve the accuracy of mineral quantification from the spectral reflectance of hyperspectral data (Koirala et al. 2019; Dixit et al. 2021; Rasti et al. 2022). These efforts also aim to tackle the nonlinear influences of the material composition (TOC effect), scattering of the reflected energy, and other processes, such as multiphoton interaction, on the spectral reflectance recorded by the sensors (Huard and Marion 2011; Chen et al. 2013; Rivard et al. 2018).

In this study, we assessed several abundance quantification approaches, including unmixing techniques, on powder samples. We measured the powder samples in the laboratory, which reduced the potential scattering energy reflected by the sensor (direct contact measurement). However, we tried to introduce machine learning to our unmixing techniques to solve the other nonlinear behaviors of the mixture that might

be caused by other factors. We applied a dimensional reduction algorithm, L2 Normalization, Principal Component Analysis (PCA), and Autoencoder (AE), to the latent space for the linear unmixing algorithm. The abundance of Quartz, Calcite, and Dolomite extracted from this study was validated with the XRD measurement from a previous study.

Dataset

Twenty-one (21) powder samples from Wadi Maqna were measured using the SR-6500 Spectroradiometer by Spectral Evolution (Fig 2). The portable spectroradiometer has UV, VNIR, and SWIR sensors with three different spectra resolutions, 1.5nm @ 700nm, 3.0nm @ 1500nm, and 3.8nm @2100nm. The reflectance spectra were later resampled into 1nm for the entire wavelength range (from 350-2500nm). The powder samples have already been measured with XRD for the abundance of quartz, calcite, and dolomite, which will be used as the ground truth in this study. Samples with an abundance of single minerals higher than 90% are considered pure minerals, and the spectra measured from these samples were used as the endmembers for the Unmixing analysis.



Figure 2—Left: SR-6500 Spectroradiometer by Spectral Evolution used in this study. Right: Powder sample measured using SR-6500.

Methodology

The methodology in this study is summarized in (Fig 3), indicating the process taken in this study from spectra measurement to the abundance quantification using various approaches. To assess the accuracy of the abundance estimation from each method, the root mean square error (RMSE) and mean absolute error (MAE) were calculated.

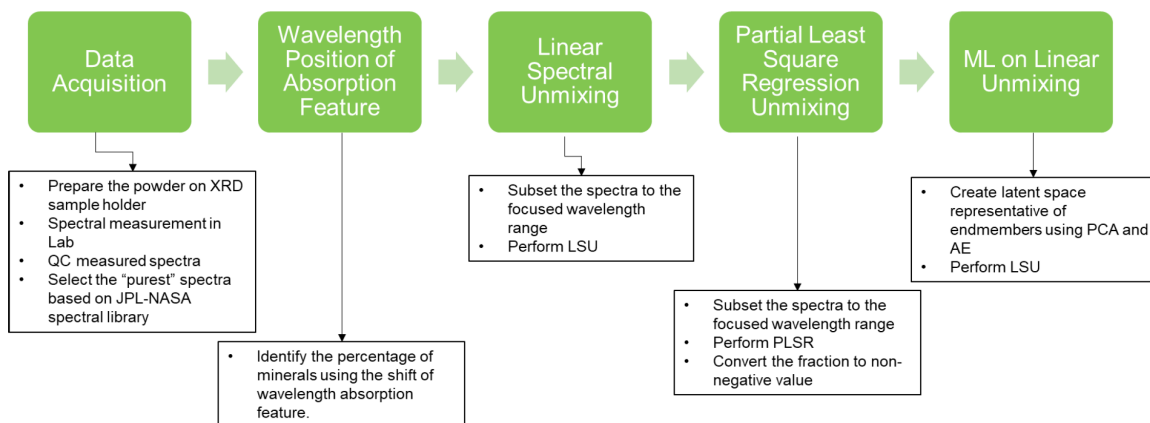


Figure 3—General workflow of this study

Wavelength Position of Absorption Feature.

Mapping the wavelength position of the absorption feature is not only useful for mineral identification from hyperspectral data (Hecker et al. 2019; van der Werff and van der Meer 2015; van Ruitenbeek et al. 2014), further analysis on the wavelength position could also be used to understand other features such as crystallinity degree, grain size and mineral mixing quantity from the reflectance spectra (Pontual, S., Merry, N., & Gamson 1997; Zaini et al. 2012; Liu et al. 2023). In carbonate rocks, many studies demonstrate the usefulness of mapping the absorption at 2320-2350nm wavelength to distinguish the domination between Calcite-Dolomite from the linear relationship of the Ca molar concentration with the longest wavelength of the absorption feature (Kurz et al. 2012; Zaini et al. 2014; McCormick et al. 2021; Gairola et al. 2024).

Linear Spectral Unmixing (LSU)

LSU is the most popular choice for most of the unmixing process in many hyperspectral studies due to its easy implementation, as it simplifies the natural mixing process and is linear with the contribution of each material to the mixed spectra recorded. The observed spectra are defined as the contribution of each endmember with additive perturbation (if it occurs), such as noise and modeling errors in the spectra (Eq. 1). The abundance is interpreted as the contribution of each endmember after given constraints, such as non-negative values of each endmember, and the summation of them is equal to 1 (Hapke 1981).

$$S_o = \sum_{i=1}^n \alpha_{ni} \rho_i + \varepsilon$$

Where:

S_o = Observed mixed spectra

n = number of endmembers

α_{ni} = fractional abundance of the i_{th} endmember

ρ_i = the spectral measurement of the i_{th} endmember

ε = noise or error added to the spectral measurement

Partial Least Squares Regression Unmixing (PLSR)

Partial Least Squares Regression (PLSR) is a statistical method that relates two datasets—a set of independent variables (predictors) and a set of dependent variables (responses)—by projecting them to a latent space that maximizes the covariance between the two. Because both the predictors (X) and responses (Y) data are projected to new spaces, the PLS family of methods is known as bilinear factor models. The PLSR constructs the estimates of linear regression between X and Y as:

$$Y = X\tilde{B} + \tilde{B}_0$$

Where Y is the response (abundance from XRD in this study), X is the reflectance from the measured hyperspectral data, $\tilde{B}$ is the regression coefficient matrix and $\tilde{B}_0$ is the intercept vector/offset computed during the fitting process on the PLSR algorithm.

Latent Space Linear Least Squares (L2 Normalization, PCA & AE)

Endmember spectra projected to the latent space using L2 (Euclidean) normalization, Principal Component Analysis, and Autoencoder. L2 normalization is the shortest distance indicated by a vector from the data, calculated by the square root of the summation of the vector's squared element magnitude. PCA and AE are the most popular dimensionality reduction algorithms in machine learning practices. PCA is a linear transformation that retains only the maximum variance from the data and projects this into the desired lower-dimensional space. On the other hand, AE is a neural network-based algorithm that learns to encode the data into lower-dimensional representation and then decodes it back to the original data. Different approaches

to defining the lower dimension representation (L2, PCA, and AE) of the endmember reflectance spectra mixture are expected to offer an alternative in estimating the abundance of hyperspectral measurements.

Focusing on the abundance of quartz, calcite, and dolomite, we also subset the reflectance spectra from 2000-2500nm, which covers the signature absorption features of the three targeted minerals. This way, both algorithms were able to improve the accuracy of endmember estimation significantly. The abundance is finally estimated using the Linear Least Squares Regression algorithm (LLS) from the projected latent space of the three methods on both full and subset spectra of the samples.

Accuracy Measurements (RMSE and MAE)

The accuracy of the unmixing techniques was assessed by comparing the estimated abundance from measured spectra with the abundance from XRD laboratory measurements using the RMSE and MAE (Eqs. 4 and 5).

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

$$MAE = \frac{\sum_{i=1}^n |y_i - \hat{y}_i|}{n}$$

Where y_i are the abundance of each mineral from XRD and $\hat{y}_i$ and x_i are the abundance from the unmixing approaches used in this study.

Results

Endmember Selection

From the twenty-one measured powder samples, five of the samples were identified as pure minerals according to the XRD and by comparing the measured spectra with the JPL-NASA spectral library of Quartz, Calcite, and Dolomite (Fig. 4). Among the three endmembers, Calcite and Dolomite have two samples with pure mineralogy. The endmember of pure calcite is indicated by the deep absorption feature at 2339nm, longer than the dolomite at 2321 nm, while the quartz is indicated by the absorption feature at 2220nm. The rest of the spectra are later grouped based on the dominant composition and balanced mixed to easily compare the abundance information from the XRD with visual interpretation using the absorption feature from the reflectance spectra.

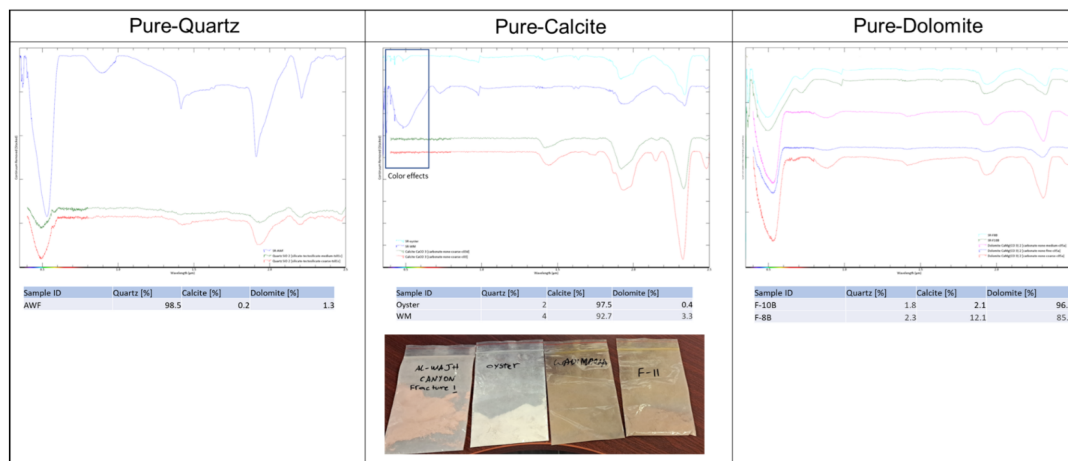


Figure 4—Reflectance spectra of the endmembers of Quartz, Calcite, and Dolomite compared to the NASA JPL Spectral library and its mineral composition from XRD.

Absorption Features Wavelength Position Mapping

The deepest absorption feature within the Calcite/Dolomite signature (2320-2350nm) of the reflectance spectra of each sample is extracted and plotted with the calculated Calcite/Dolomite ratio and the actual calcite percentage to quantify the relative composition of quartz, calcite, and dolomite in Wadi Maqna samples.

Sample ID	XRD			Calc/Dolo Ratio	Wavelength Position
	Quartz	Calcite	Dolomite		
AWF	98.5	0.2	1.3	-	-
F1	1.6	45.6	52.8	0.463414634	2330
F2	54.1	5.9	39.9	0.128820961	2321
F3	50.7	49.1	0.2	0.995943205	2339
F4	2.6	16	81.4	0.164271047	2322
F5	0.7	74.4	25	0.748490946	2336
F6	54.3	42.8	2.9	0.93654267	2338
F7	40.9	58.1	1	0.983079526	2338
F8-A	2.3	7.3	90.4	0.074718526	2325
F8-B	2.3	12.1	85.6	0.123848516	2321
F-9	23.6	76.3	0.1	0.998691099	2336
F-10-A	3	63.5	33.5	0.654639175	2334
F-10-B	1.8	2.1	96.1	0.021384929	2322
F-11	3.9	44.4	51.7	0.46201873	2331
OYS	2	97.5	0.4	0.995914198	2337
SB-1	44	55.8	0.1	0.998211091	2340
SB-2	58.1	30	12	0.714285714	2337
SB-3	70.6	28.2	1.2	0.959183673	2338
SB-4	68	32	0	1	2339
WM	4	92.7	3.3	0.965625	2338

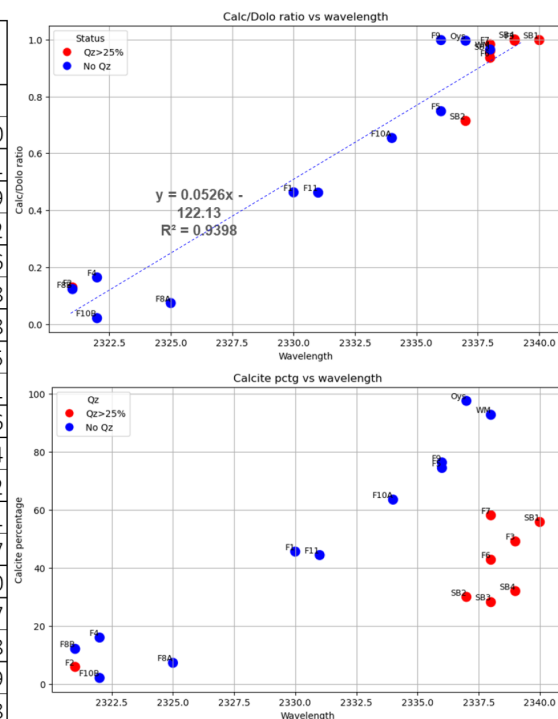


Figure 5—Mineral composition of the sample from XRD and plot of wavelength location of absorption feature with Calcite/Dolomite ratio and calcite percentages.

In the calcite/dolomite ratio vs absorption feature plot, the correlations are very high for all the samples except for SR-F2 (Fig. 6). According to XRD measurement, the sample contained 54% of quartz, 5.9% of calcite, and 39.9% of dolomite. A similar result is shown by the cross plot between the calcite percentage vs absorption feature wavelength, except for the SR-F2 sample. Additionally, the shift influenced by the presence of the quartz was highlighted vividly at the isolated cluster of absorption features in the longer wavelength > 2337.5 nm, except for SR-SB2.

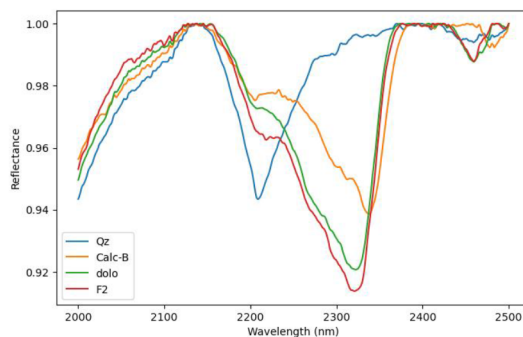


Figure 6—Reflectance spectra of SR-F2 are likely to have higher dolomite compared to quartz. Closely similar to the spectra of dolomite's endmember.

LSU and PLSR

The results of LSU and PLSR on the Wadi Maqna samples are summarized in Table 1. Unmixing methods could not estimate the abundance of each mineralogy using the full reflectance spectra accurately. The average accuracy scores for the LSU on full spectra are 34,6 and 30,5, ranging from 9,2 to 68,8 and 7,7 to 64,7 for RMSE and MAE, respectively. PLSR performs slightly worse with an average of 35,4 and 31,3 with a range of 7,8 to 68,8 and 6,4 to 64,7 for RMSE and MAE. The worst predicted sample was F6, where both algorithms completely misinterpreted the spectra, stating that the sample is 100% dolomite, while in XRD analysis, there is a mixture of almost 50:50 of quartz and calcite.

Table 1—Summarized unmixing results from LSU and PLSR with the RMSE and MAE scores for every sample.

Sample ID	XRD			Linear Unmix (Full)			RMSE Score	MAE Score	Linear Unmix			RMSE Score	MAE Score	PLSR(Full)			RMSE Score	MAE Score	PLSR			RMSE Score	MAE Score
	Quartz	Calcite	Dolomite	Quartz	Calcite	Dolomite			Quartz	Calcite	Dolomite			Quartz	Calcite	Dolomite			Quartz	Calcite	Dolomite		
AWF	98.5	0.2	1.3																				
F1	1.6	45.6	52.8	5.7	62.2	32.1	15.5	13.8	0	51	49	3.9	3.6	3.4	66.4	30.2	17.8	15.1	0	50.5	49.5	3.5	3.3
F2	54.1	5.9	39.9	20.1	66	13.9			0	1	99			18.9	67.9	13.2			0	0	100		
F3	50.7	49.1	0.2	41.1	28.4	30.5	21.9	20.2	56.9	43.1	0	5.0	4.1	39.3	31	29.7	21.0	19.7	58.9	37.3	3.8	8.6	16.4
F4	2.6	16	81.4	19.9	38	42.1	27.9	26.2	0	20.7	69.3	7.6	6.5	18.7	39.8	41.5	28.4	26.6	0	31.4	68.6	11.7	19.5
F5	0.7	74.4	25	11.5	62.8	25.7	9.2	7.7	1.7	74.2	24.1	0.8	0.7	10.2	64.8	25	7.8	6.4	1.8	70.6	27.6	2.7	5.3
F6	54.3	42.8	2.9	0	0	100	68.8	64.7	52.8	32.2	15	9.3	8.1	0	0	100	68.8	64.7	53	26.8	20.2	13.6	50.0
F7	40.9	58.1	1	7.2	0	92.8	65.7	61.2	59.4	40.8	0	14.6	12.3	3.6	0	96.4	68.0	63.6	62.6	37.4	0	17.3	51.5
F8-A	2.3	7.3	90.4																				
F8-B	2.3	12.1	85.6																				
F-9	23.6	76.3	0.1	23.8	23.4	52.8	43.1	35.3	43.2	56.8	0	16.0	13.1	22.7	25.1	52.2	42.2	34.7	43.7	56.3	0	16.4	24.4
F-10-A	3	63.5	33.5	7.7	15.8	76.5	37.2	31.8	10.3	65.4	24.3	6.9	6.1	6.9	17	76.1	36.5	31.0	10.4	64.3	25.3	6.4	18.5
F-10-B	1.8	2.1	96.1																				
F-11	3.9	44.4	51.7	10.7	82.7	6.6	34.4	30.1	10.6	44.7	44.7	5.6	4.7	11.5	81.2	7.3	33.6	29.6	10.9	41.1	48	5.0	17.2
OYS	2	97.5	0.4																				
SB-1	44	55.8	0.1	5.7	91.3	3	30.2	25.6	14.9	85.1	0	23.8	19.5	4.6	93.2	2.2	31.4	26.3	18	82	0	21.3	22.6
SB-2	58.1	30	12	33.8	44.5	21.7	17.3	16.2	49.7	50.6	0	14.6	13.7	31.8	47.8	20.5	19.0	17.5	52.6	46.4	1	11.8	20.7
SB-3	70.6	28.2	1.2	24.7	75.3	0	38.0	31.4	70.8	29.2	0	0.9	0.8	21.9	21.2	56.6	42.8	37.0	71.9	25.6	2.5	1.8	35.1
SB-4	68	32	0	18	82	0	40.8	33.3	54	46	0	11.4	9.3	15.6	84.4	0	42.8	34.9	56.9	43.1	0	9.1	21.2
WM	4	92.7	3.3																				

The average RMSE and MAE scores from both algorithms decrease to <10%, except for the MAE score from the PLSR. Despite the overall good results, some samples miscalculated the abundance quite significantly, such as SB-1 and F-9. As explained in the previous section, the F-2 sample also indicates pure dolomite, though this is not completely correct, as we still can see the effect of quartz in the reflectance spectra (Fig. 6). Therefore, the RMSE and MAE scores from sample F2 were not calculated.

Latent Space LLS Unmixing

While L2 normalization (L2-LLS) and autoencoder (AE-LLS) were able to estimate the abundance of the powder samples, PCA failed to perform the unmixing of the samples in many of the samples. Further inspection is required to understand this issue. Therefore, the estimation result from PCA is excluded from further analysis, and the focus is on the result of L2 and AE-LLS.

Similar to the LSU and PLSR unmixing, latent space unmixing using L2-LLS and autoencoder AE-LLS could not estimate the correct abundance of the powder samples using the full spectra (table. 2). The average RMSE and MAE for L2-LLS are 26,9 and 22,85 for the full spectra and 8,4 and 6,8 for the subset spectra. The overestimation of calcite abundance in sample SB-2 is also found from the calculation using this method. While the other three unmixing methods have good results in the subset spectra, AE-LLS failed to duplicate the improved estimation.

Table 2—Summarized unmixing results from L2-LLS and AE-LLS with the RMSE and MAE scores for every sample

Sample ID	XRD			L2 LLS (Full)			RMSE		MAE		L2 LLS			RMSE		MAE		AE LLS (Full)			RMSE		MAE		AE LLS			RMSE		MAE		PCA LLS (Full)			PCA LLS		
	Quartz	Calcite	Dolomite	Quartz	Calcite	Dolomite	Score	Score	Quartz	Calcite	Dolomite	Score	Score	Quartz	Calcite	Dolomite	Score	Score	Quartz	Calcite	Dolomite	Score	Score	Quartz	Calcite	Dolomite	Score	Score	Quartz	Calcite	Dolomite	Score	Score				
F1	1.6	45.6	52.8	5.7	62.4	31.8	15.7	14.0	0	54.5	45.5	6.7	5.9	42.6	32.4	24.8	29.7	22.8	33.5	33.9	32.5	27.4	21.3	19.4	17.6	62.9	0	9.5	90.5	0	0	0	0	0			
F2	54.1	5.9	39.9	6.3	93.7	0			0	0	1	0	38.2	61.8						33.5	33.9	32.5			nan	nan	nan	nan	nan	nan	nan	nan	nan				
F3	50.7	49.1	0.2	40.8	28.7	30.4	21.8	20.2	56.3	43.7	0	4.5	3.7	39.1	27.8	32.9	23.5	32.6	70.3	3.2	26.5	21.9	30.6	nan	nan	nan	nan	nan	15.3	24.3	74.1	0	0				
F4	2.6	16	81.4	19.8	38.4	41.7	28.1	26.4	0	12.9	87.1	4.0	3.8	45.5	22.4	32.1	37.9	47.9	65.2	7.2	27.6	32.9	41.7	nan	nan	nan	nan	nan	nan	nan	nan	nan					
F5	0.7	74.4	25	11.4	63.2	25.4	8.9	7.4	1.8	74	24.2	0.8	0.8	26.8	38.4	34.8	26.3	37.0	64.7	76.1	27.2	24.0	22.6	2.1	0	97.9	0	11.5	88.5	0	0	0	0				
F6	54.3	42.8	2.9	0	0	100	68.8	64.7	52.8	32.2	15	9.3	8.1	0	0	100	68.8	64.7	53	26.8	20.2	13.6	50.0	0	0	100	68.8	64.7	53	26.8	20.2	13.6	50.0				
F7	40.9	58.1	1	7.2	0	92.8	65.7	61.2	59.4	40.8	0	14.6	12.3	3.6	0	96.4	68.0	63.6	62.6	37.4	0	17.3	51.5														
F-9	23.6	76.3	0.1	11.9	55.8	32.3	23.1	21.5	33.6	66.4	0	8.1	6.7	39	32.9	27.1	30.8	31.3	33.5	33.9	32.5	28.6	28.2	36.4	33.9	29.7	4.4	0	95.6	4.4	0	0					
F-10-A	3	63.5	33.5	1.5	35.3	63.2	23.7	19.8	10.3	65.4	24.2	6.9	6.2	35.7	33.5	30.8	25.7	24.5	33.5	33.9	32.5	21.8	20.4	nan	nan	nan	nan	nan	0	2.6	97.4	0					
F-11	3.9	44.4	51.7	20.5	42.1	37.4	12.7	11.1	10.6	44.5	44.9	5.5	4.5	31.3	34.5	34.2	19.6	44.6	66.5	6.2	27.3	18.3	41.7	0	100	0	0	56.4	43.6	0	0	0					
SB-1	44	55.8	0.1	5.7	91.4	2.9	30.2	25.6	17.3	82.7	0	21.9	17.9	25.2	39.6	35.2	24.8	34.6	65.4	7	27.6	23.4	32.6	0	0	100	nan	nan	nan	nan	nan	nan					
SB-2	58.1	30	12	33.5	45.2	21.3	17.5	16.4	49.1	50.9	0	14.9	14.0	33.5	32.6	33.5	19.1	18.8	69.5	3.8	26.7	16.4	17.4	25.8	0	74.1	16.9	83.1	0	0	0	0					
SB-3	70.6	28.2	1.2	27	72	0	36.1	29.9	70.6	29.4	0	1.0	0.8	32.1	33.7	34.2	29.4	20.9	71.8	1	26.3	25.7	17.5	0	0	100	1.1	4	94.9	0	0	0					
SB-4	68	32	0	22.1	77.9	0	37.5	30.6	51.5	48.5	0	13.5	11.0	24.4	40.3	35.3	32.7	22.6	70.2	3.3	26.5	29.1	19.1	0	0	100	27.8	72.2	0	0	0	0					

Comparison of Conventional vs Deep Learning Embedded Unmixing

The unmixing performed on the subset dataset always outperformed the unmixing on the full dataset, except in the AE-LLS algorithm. In general, the L2-LLS produces the best result among other unmixing methods. The RMSE and MAE scores of the L2-LLS for the full and subset spectra are superior to LSU, PLSR, and AE-LLS (Table 3).

Table 3—Overall RMSE and MAE score of unmixing techniques on full and subset data

Methods	LSU	LSU subset	PLSR	PLSR subset	L2-LLS	L2-LLS Subset	AE-LLS	AE-LLS Subset	PCA-LLS	PCA-LLS Subset
RMSE Score	34.6	9.3	35.4	9.9	26.9	8.4	27.6	31.1	n.a.	n.a.
MAE Score	30.6	7.9	31.3	23.5	22.9	6.9	24.1	28.1	n.a.	n.a.

Discussions

Among all the samples, only one sample does not match the XRD measurement, sample F2, which was analyzed as a mixture of quartz and dolomite. However, the reflectance spectra of F2 show very similar spectra to pure dolomite (F8-B), indicated by the absorption feature in the shorter wavelength at 2321nm (Zaini et al. 2012) with only a shallow feature at 2220 nm. Therefore, we exclude this sample from further abundance estimation comparison, which will automatically give the worst RMSE and MAE scores.

While the two spectra of the Dolomite's endmembers (F-8B and F-10B) are identical, there is a variation in the spectra of the Calcite's endmember within the visible light wavelength range. This might be caused by the occurrence of iron minerals in a pure calcite sample with code Wadi Maqna (WM) compared to the oyster, which appeared in a darker color, supported by the presence of an absorption feature of iron-bearing composition at the VNIR wavelength range, specifically 500nm, 750nm, and 1000nm (Baranoski et al. 2014) of the sample with the code WM that was ignored in the XRD measurement (Fig. 4).

In many samples, the PCA transformed spectra fell outside the Euclidean distance of the endmembers (Fig. 7). Thus, the abundance of each mineral in the recorded samples cannot be estimated. Due to the frequent occurrence of this case, the RMSE and MAE of PCA unmixing were not calculated. Observing the PCA plot, the contribution of other minerals might indeed appear in the measured samples. Thus, trying this approach again with the additional endmembers from the previously mentioned complete XRD result might solve this issue.

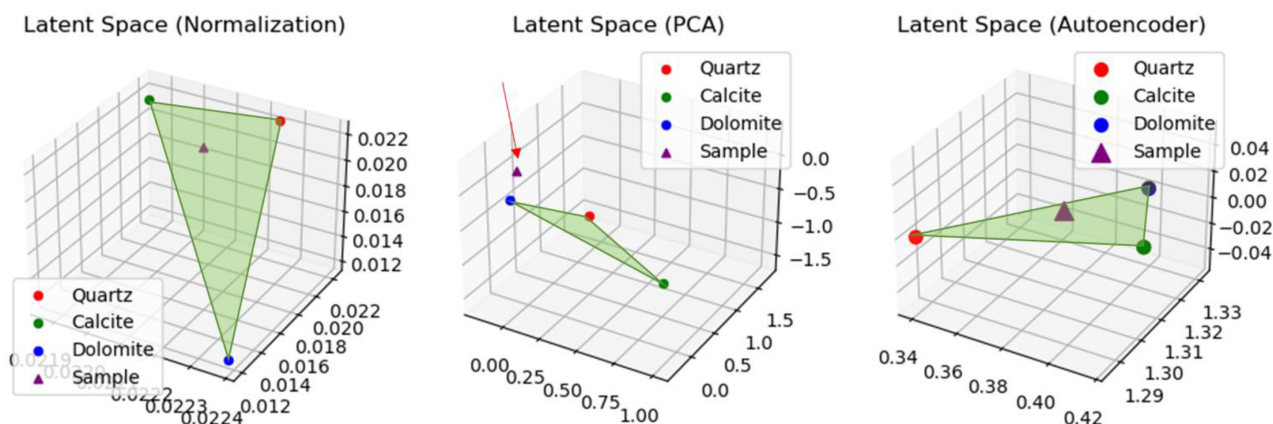


Figure 7—Latent space representation of the endmember spectra and measured sample in L2 normalization, PCA, and autoencoder.

Among all tested algorithms, the AE-LLS unmixing technique outperforms others when applied on full spectra data, based on RMSE and MAE scores, compared to subset data. However, the abundance extracted from AE-LLS is still unreliable, as none of the abundance estimation matches with the XRD data despite the better RMSE and MAE scores. The RMSE and MSE from the subset spectra of AE-LLS produced the worst accuracy and completely estimated the wrong distribution in many samples.

AE performance in general is sensitive to the choice of hyperparameters, including the randomness of the latent projection (Krček and Perin 2024). This may cause the latent space projection not to align directly with the abundance of the endmember. For this reason, we built a simple AE architecture and only constrained the random seed gathered from the best-performing model used in the test data, assuming the best-performing model has a non-linear relationship with the abundances. Referring to a previous study that successfully used AE for hyperspectral unmixing (Palsson et al. 2022), we assume that LLS might not be the best choice in estimating the abundance of endmembers from AE latent space, especially for a much smaller dataset that was used in this study. Previous studies demonstrated the use of AE for hyperspectral unmixing by gathering the endmember from a large dataset of spectra from hyperspectral imagery, they estimated the abundance by training the decoder's weight set of their autoencoder after the endmember extraction.

Despite the good accuracy, it is important to note that the abundance estimation on the full spectra by L2-LLS is not without any shortcomings. The better score is merely because of the fewer errors instead of the correctness to the actual value estimated by the unmixing method, since both are wrongly estimating the abundance in most of the samples. The best-performing unmixing algorithms in this study are LSU and L2-LLS unmixing, where the deep-embedded algorithm (L2-LLS) outperforms the conventional (LSU) approach by a small margin.

The poor accuracy given by the unmixing on full spectra is caused by the different colors of the samples that were selected as the endmembers, as mentioned earlier. Another issue is the lack of an absorption feature identifier of quartz in VNIR and SWIR. The absorption feature used for the pure quartz, at 2200nm, does not originally come from the Si-O-Si bond, which is located in the LWIR (Hecker et al. 2019), and it normally comes from the Al-OH bending, corresponding to the existence of clay minerals. Therefore, complete mineralogy composition from XRD should be explored and introduce more endmembers from the actual mineral composition of the sample to both unmixing methods.

Conclusions

In this study, we compared qualitative and quantitative mineral composition estimation methods from hyperspectral reflectance spectra measured from powdered samples. Qualitative composition using wavelength mapping in Calcite/Dolomite signature wavelength range (2320-2350nm) was able to determine the relative composition of calcite and dolomite, as demonstrated by many previous studies. In addition to that, our result is also able to distinguish the influence of the significant presence of quartz by the shift of the absorption feature to the longer wavelength.

Amongst all the unmixing techniques used in this study, Linear Spectral Unmixing and L2 Normalized-LLS are better compared to others. While both are linear unmixing techniques, we conclude that the abundance of minerals in our samples is indeed linear with the spectra contribution from the endmembers. The effect of non-linearity is not present in the sample, nor the expected TOC effect and scattering energy reflected, nor the other processes such as multiphoton interaction.

Projecting the spectra into latent space consistently reduces the MAE and RMSE scores despite its incorrect abundance estimation in the full spectra. Applying these techniques to complicated samples (more mixture or non-linear distribution) might be useful, as this method could deal with non-linear relationships as demonstrated in many other cases, including hyperspectral unmixing on imagery.

Given the high accuracy ($R^2 = 0.94$ between absorption features position vs calcite-dolomite ratio and $RMSE < 10$ for LSU and L2-LLS unmixing algorithms) in estimating rock mineralogy from VNIR and

SWIR hyperspectral spectra, we recommend using a portable spectroradiometer for reliable, non-destructive preliminary mineralogy analysis of rock samples or outcrops. In addition, future work should focus on investigating the failure of autoencoder and PCA in performing unmixing using this sample by providing more detailed mineralogical information from XRD to improve the accuracy of abundance estimation of the rocks.

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